Kim 2020 LUAD — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity **GO**

13/14 samples (≥42 cells) intermix on one continuous manifold; radius-graph finds 0 detached islands. Harmony θ=5 (mixing plateaus by θ=5); both compartments span multiple patients.

Gate 2 — module geography **GO** (non-branching)

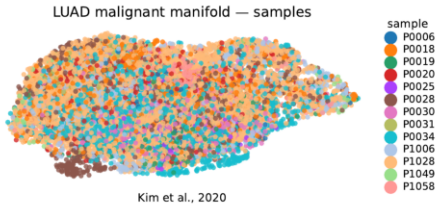
Gast/pSom near-orthogonal (r=−0.11), pSom broadly dominant over the body; Crest anti-correlated (r=−0.55 / −0.41), rising to a terminal readout (ρ=+0.48; crest-high decile mean pt=0.81). Gast declines along pseudotime (ρ=−0.55) — one monotone Gast/pSom → neural-crest axis, no second arm.

Gate 3 — hub mutual exclusivity **NA**

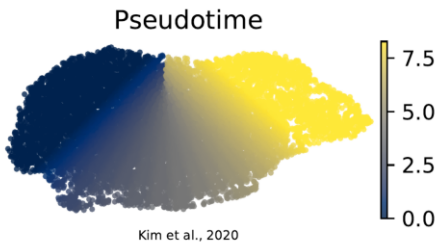
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test.

STEP FIGURES

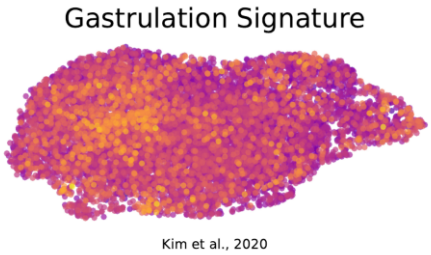
Gate 1: integration



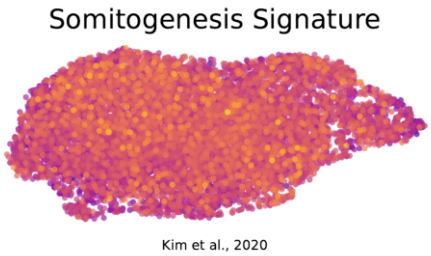
Trajectory / root



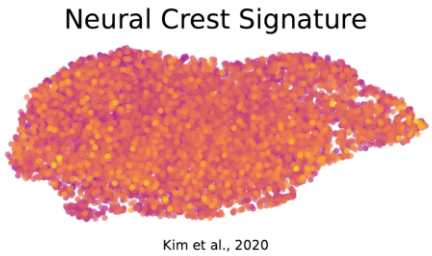
Gast



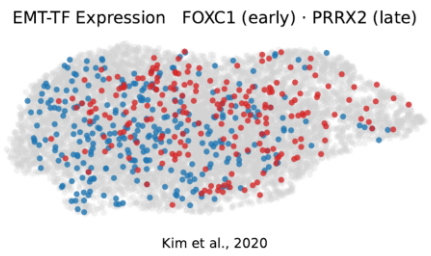
pSom



Crest



EMT-TFs



Dataset. Lung Adenocarcinoma (Kim et al., 2020), 10x — 14 patients / 32,493 cells (master scope); malignant only: 10,538 load, 10,512 across 13 samples (≥42) enter preprocessing. The whole malignant manifold (no lobe scoping) enters the trajectory: Embryonic 9,315 (6% cycling) + Cycling 1,197 (61%).

Verdict. ✓ Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Kim's LUAD malignant manifold is a single monotone Gast/pSom → terminal-neural-crest gradient, matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis spans both proliferation lobes (orthogonal to cell cycle; per-lobe module means near-identical, Crest-high decile split 52/48), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). Caveat: pseudotime co- varies with cycling (ρ=+0.41). EMT-TFs FOXC1 (early/root) → PRRX2 (late/terminus); canonical SNAI1/ZEB1/TWIST1 near the detection floor, end-separation weak (Δpt=0.21).

Parameter choices

Min cells / sample:	≥42 (13/14 kept; gap 70→26, ratio 2.7)	**Integration:**	Harmony θ=5 on 50-PC; UMAP spread=1.3
Scope:	whole manifold (no lobe scoping)	**Trajectory:**	scFates curve N=20, λ=0.3, μ=100
Root node:	tip 0 = Gast/pSom-high, Crest-low	**Compartments:**	Embryonic 9,315 / Cycling 1,197